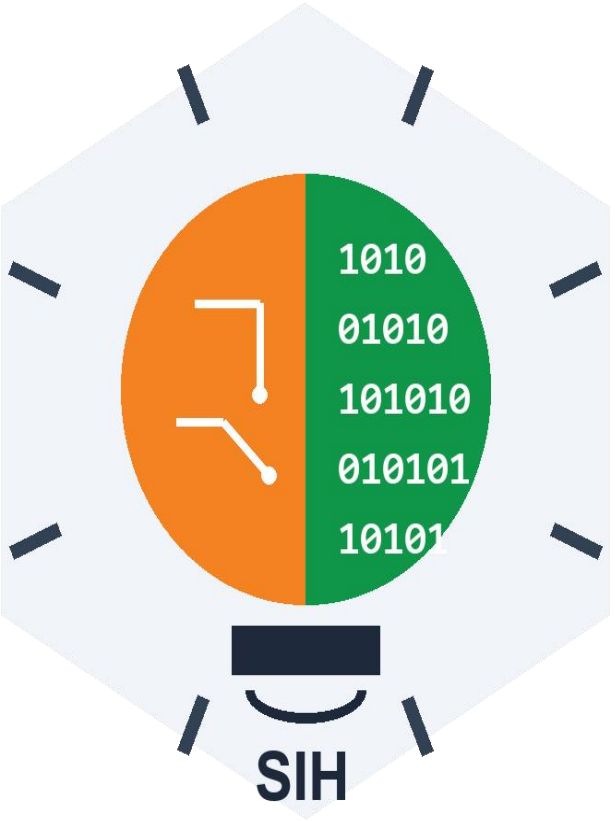
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- **Problem Statement ID –** SIH26103
- **Problem Statement Title -** Integrated Government Project Monitoring & Intelligence Platform
- **Theme-** Infrastructure & Governance
- **PS Category-** Software
- **Team ID-** SIH2026-VX103
- **Team Name -** Team VeriX
- **Project Name -** SetuX ("From Reported Progress to Verified Progress")



Team VeriX

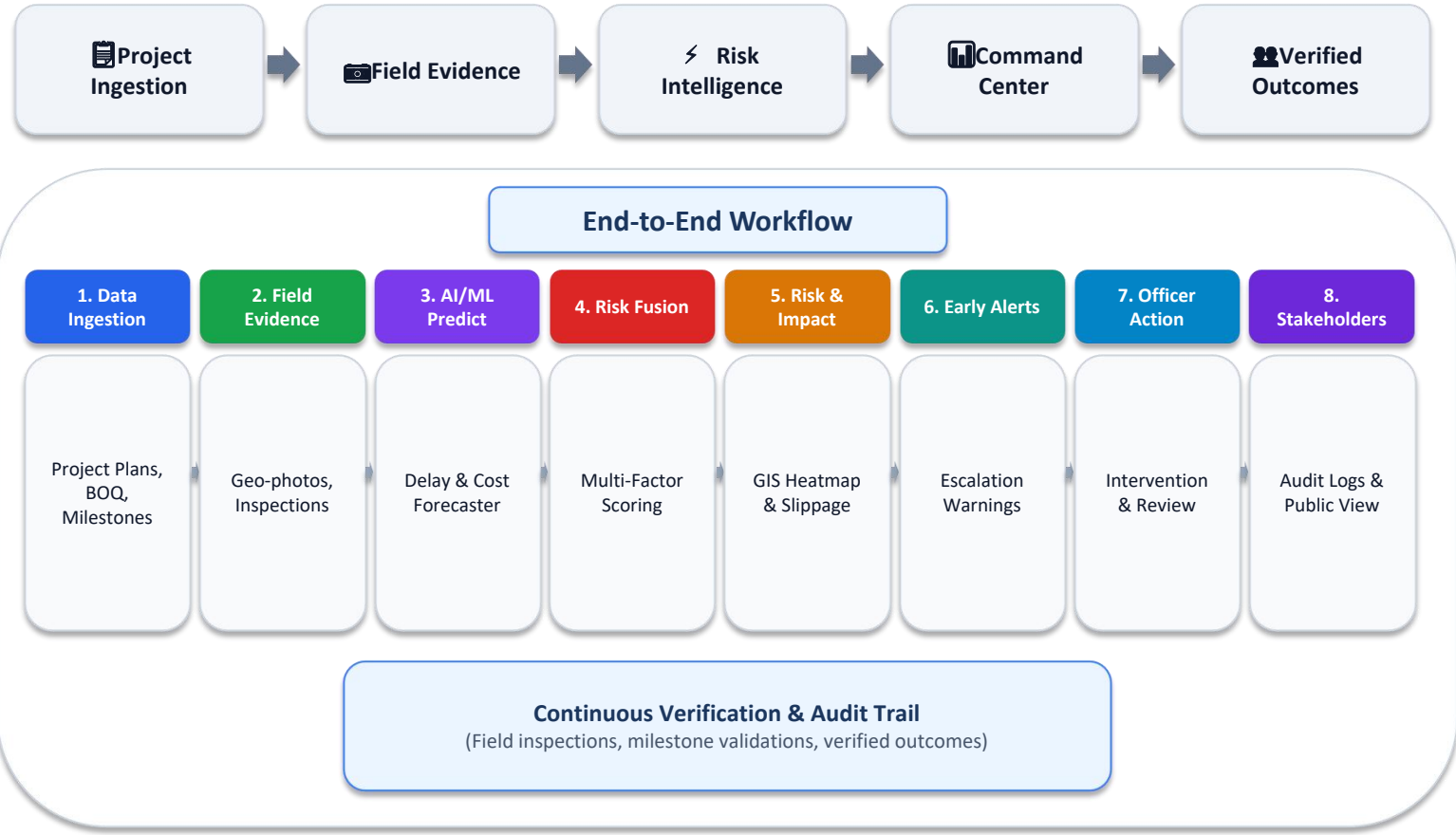
SetuX

Bridging Infrastructure Monitoring to Verified Progress

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Proposed Solution & Approach

1. **Project Monitoring:** Centralized tracking of physical, financial, and milestone status across all projects.
2. **Progress & Evidence:** Multi-source progress backed by geo-tagged photos, documents & inspection logs.
3. **Risk Intelligence:** Multi-factor scoring evaluating milestone delays, cost escalation & vendor risk.
4. **Predictive Analytics:** Machine learning models forecasting potential cost overruns and schedule delays.
5. **Early Warnings:** Proactive alerts and trigger thresholds notifying officers before slippage escalates.
6. **Executive Reporting:** Automated, audit-ready briefing reports and executive dashboards for quick decisions.



Hosted Application
Click here

GitHub Repository
Click Here

Team VeriX

TECHNICAL APPROACH

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Technology Used

Data Sources
- Project Plans / DPR
- Milestone Schedules
- Financials / BOQ
- Geo-coordinates

Data Processing
- Python / FastAPI
- Pydantic Schemas
- SQLAlchemy ORM
- NumPy / Pandas

AI/ML Model
- scikit-learn Engine
- Cost Overrun Regr.
- Delay Risk Classifier
- Rule Risk Indicators

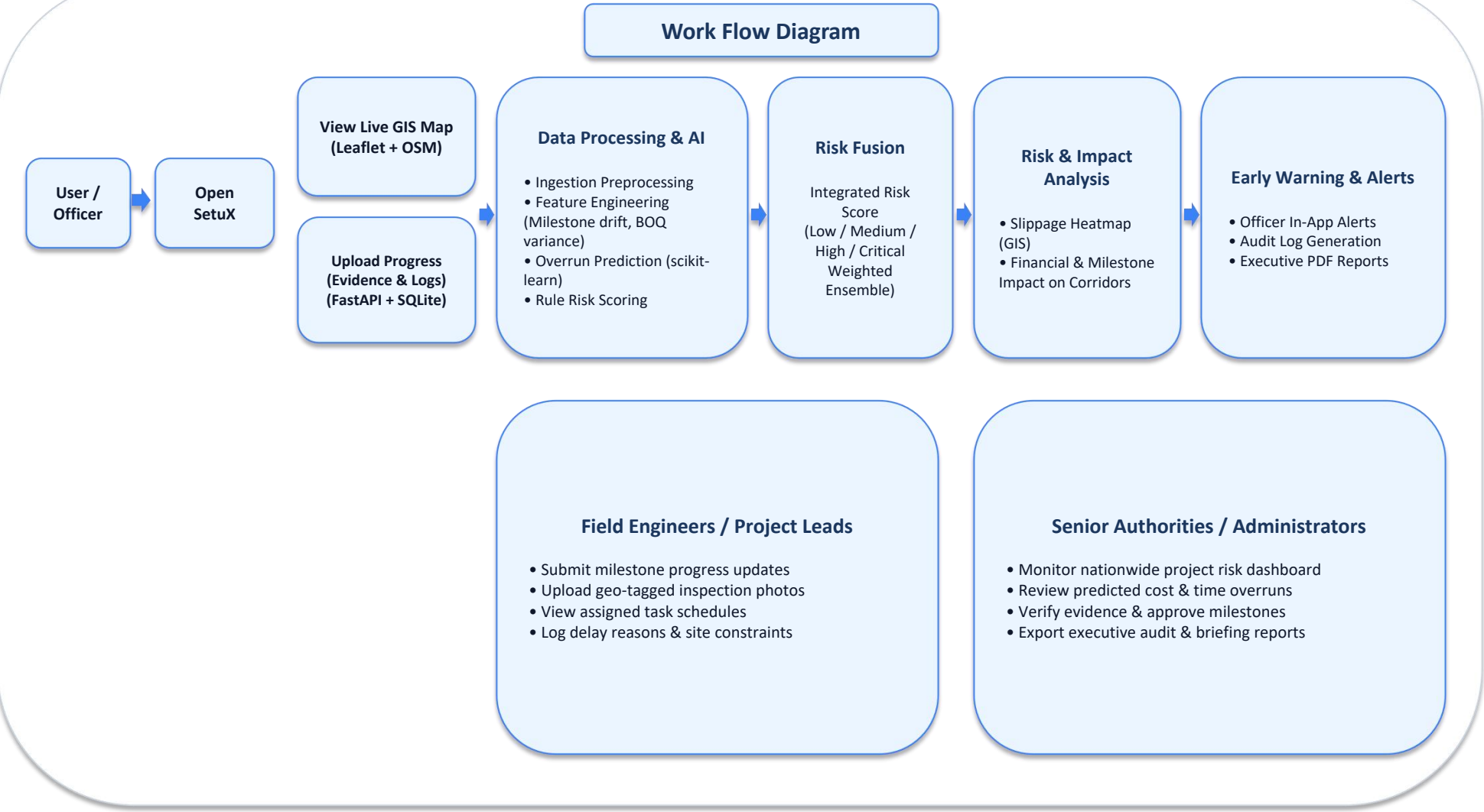
Database
- SQLite (MVP)
- PostgreSQL Schema
- Audit Event Logs

Maps & GIS
- Leaflet / OSM
- GeoJSON Pins & Poly
- Spatial Risk Heatmap

User Interface
- React 18 / Vite
- Tailwind CSS
- Recharts Analytics
- Lucide Icons

Deployment
- Uvicorn ASGI
- RESTful Architecture
- GitHub / Docker

Work Flow Diagram



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FEASIBILITY AND VIABILITY

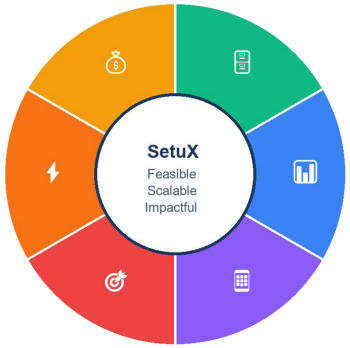
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Feasibility of the Idea

Available Data Sources
Standard DPRs, milestone schedules, financial allocations, and field progress logs.

Technically Viable
Proven React + FastAPI stack with scikit-learn predictive modeling & Leaflet GIS.

Field Practical
Lightweight web architecture with responsive UI & offline-tolerant data entry.



Cost-Effective
Built on modern open-source stack, eliminating expensive proprietary licensing.

Scalable Solution
Modular REST API architecture easily expanding from local works to national portfolios.

Actionable Insights
Intuitive risk scores, time/cost overrun forecasts, and spatial project maps.

Challenges Faced & Strategies to Overcome

Challenge	Strategy
Fragmented Project Data in Silos Multi-source disparate spreadsheets delay decision visibility.	**Unified Ingestion & Dashboard** Standardized REST API ingestion creates a single pane of glass monitoring system.
Subjective Progress Reporting Field claims without evidence lead to unaccounted project delays.	**Evidence-Backed Verification** Geo-tagged photos, BOQ logs & inspection checkpoints validate field progress.
Unforeseen Cost & Time Overruns Delays discovered only after critical budget slippages occur.	**ML Risk Intelligence Engine** scikit-learn predictive models forecast overrun probability & schedule drift.
Complex Multi-Site Spread Difficult to track geographical progress across highway corridors.	**GIS Interactive Mapping** Leaflet + OpenStreetMap maps all projects with visual risk heatmaps.
Administrative Reporting Burden Manual compilation of executive summaries wastes officer time.	**Automated Executive Reports** Instant PDF generation with structured KPI summaries & audit logs.

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BENEFIT AND IMPACT

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Unified Project Monitoring
Centralized real-time visibility across physical, financial, and milestone metrics.

01

Evidence-Supported Progress
Geo-tagged photos, documents, and inspection logs validate claimed progress.

02

Predictive Risk Intelligence
Multi-factor risk engine detects schedule slippages and cost anomalies early.

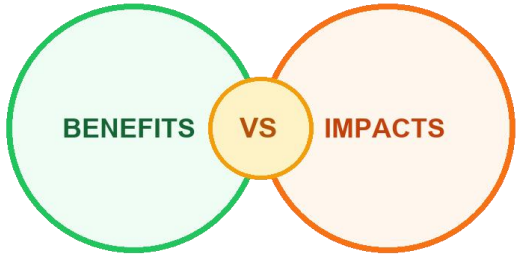
03

Geospatial Infrastructure Mapping
Interactive GIS tracking enables regional and corridor-level oversight.

04

Audit-Ready Reporting
Automated executive briefings and complete audit trails streamline compliance.

05



01

Timely Officer Intervention
Proactive alerts enable administrative action before minor delays become critical overruns.

02

Enhanced Accountability
Verifiable progress submissions reduce discrepancy between reported and actual field status.

03

Optimized Resource Allocation
High-risk projects are highlighted for targeted intervention and budget management.

04

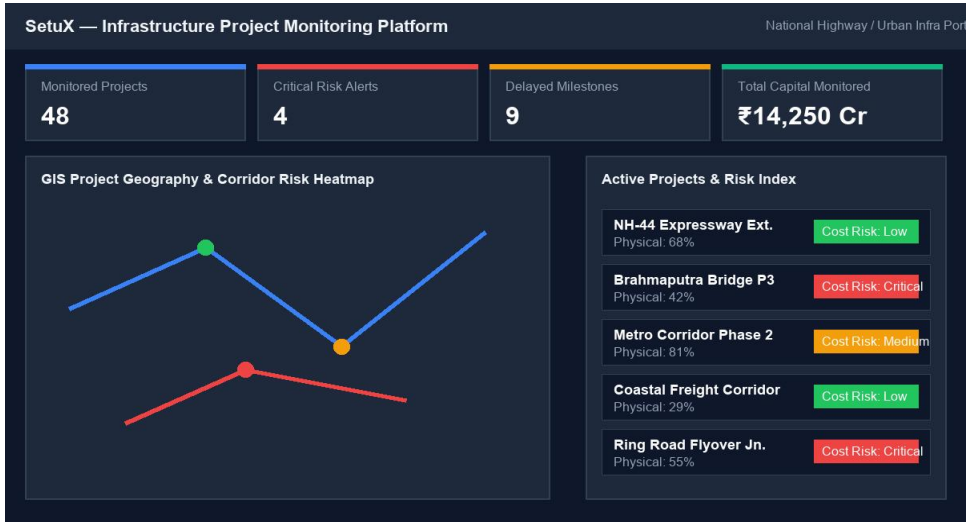
Improved Governance Visibility
Standardized monitoring framework ensures consistent oversight across departments.

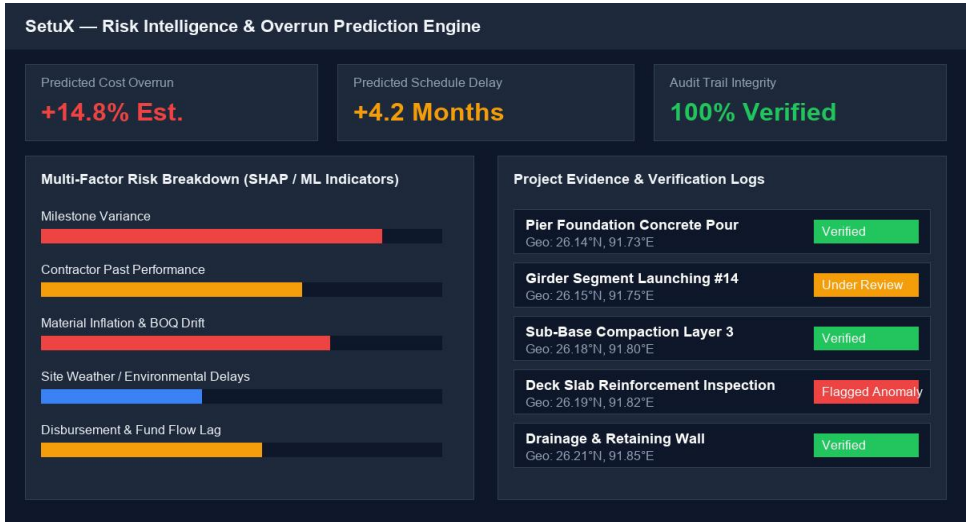
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RESEARCH AND REFERENCES

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  - Infrastructure and Project Monitoring Division (IPMD) — National project monitoring norms & flash reporting system.
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- **Government of India — PM GatiShakti National Master Plan**
  - Multi-modal infrastructure connectivity, spatial planning and institutional project monitoring framework.
  https://gati-shakti.gov.in/
- **Project Monitoring Group (PMG) / DPIIT**
  - Institutional mechanism for fast-tracking resolution of issues and milestone monitoring in mega projects.
  https://pmg.dpiit.gov.in/
- **scikit-learn: Machine Learning in Python — 2026**
  - Pedregosa et al., JMLR. Applied for infrastructure cost overrun regression and schedule delay risk classification.
  https://scikit-learn.org/stable/
- **FastAPI & High-Performance Asynchronous Python Framework**
  - Production-ready REST API architecture with Pydantic schema validation for robust government data pipelines.
  https://fastapi.tiangolo.com/
- **Leaflet & OpenStreetMap Open Geospatial Standards**
  - Open-source interactive GIS visualization for project corridors, geo-tagged checkpoints and spatial heatmaps.
  https://leafletjs.com/





DASHBOARD